

Agentic Stability Framework

ASF Specification and Whitepaper v1.1

A force-based enterprise architecture framework for governing autonomous AI systems

Publication status: Versioned release. Retains the v1.0 core model; adds non-compensatory rules and assessment-boundary clarifications. Released openly for practitioner adoption, field feedback, and future versioned improvements.

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Core Thesis

Agentic maturity is not the ability to increase autonomy. Agentic maturity is the ability to increase autonomy while maintaining stability.

$$\text{Stability Margin Index (SMI)} = \text{Enterprise Gravity Index (EGI)} - \text{Agentic Force Index (AFI)}$$

Field	Value
Version	v1.1
Supersedes	v1.0 Initial Public Release
Audience	CIO, CISO, CTO, enterprise architects, platform teams, AI governance boards
Scope	Agentic-system-level stability assessment for autonomous and semi-autonomous AI systems
Status	Versioned release; feedback and field validation welcome for future versions.
Companion artifact	ASF Assessment Workbook v1.1

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1. Executive Summary

Enterprises are moving from assistive AI to agentic AI. Assistive systems generate information. Agentic systems can reason, plan, call tools, access memory, coordinate with other agents, and act on enterprise systems.

This shift changes the architecture question. It is no longer enough to ask whether an AI output is accurate. The more important question is whether the enterprise can govern what the AI system is allowed to do.

The Agentic Stability Framework (ASF) provides an agentic-system-level model for assessing whether autonomous capability is proportional to enterprise control strength. ASF does not claim to prove that an agent is safe. It indicates whether the governance environment is strong enough to justify the agent's current or proposed level of autonomy.

Primary Definition

ASF is an enterprise architecture assessment model for determining whether the governance environment around an autonomous AI agent is strong enough to contain the operational force created by that agent in a specific business context.

Version 1.1 retains the v1.0 core model unchanged. AFI, EGI, SMI, and the seven force and seven gravity dimensions are identical. v1.1 adds non-compensatory rules to the orbit determination, closing two cases in which a compensatory average could return a favourable classification for an agent that should not receive one, and clarifies the assessment boundary for products, multi-agent workflows, and controls the assessing organization does not operate. Section 22 records the full change history.

ASF Index	Purpose	Question Answered
Agentic Force Index (AFI)	Measures operational pressure created by autonomy.	How much force does this agent create?
Enterprise Gravity Index (EGI)	Measures governance and control strength around the agent.	How much control exists around this agent?
Stability Margin Index (SMI)	Compares gravity against force.	Is governance proportional to autonomy?



ASF treats SMI as a directional architecture indicator, not a proof of safety.

Figure 1. ASF core model.

2. Introduction: Autonomy Is Outpacing Governance

The first wave of enterprise generative AI was mostly assistive: summarization, drafting, search, code suggestions, and knowledge access. These capabilities introduced important concerns around accuracy, privacy, security, and responsible use, but the primary output was usually informational.

The next wave is agentic. Agents can interpret goals, create plans, select tools, call APIs, retrieve enterprise context, maintain memory, collaborate with other agents, and trigger real business or technology actions.

This creates action risk. A chatbot that gives a wrong answer creates a quality problem. An agent that deploys the wrong workload, grants the wrong access, updates the wrong customer record, invokes the wrong API, or triggers the wrong financial workflow creates an operational stability problem.

The central enterprise challenge is that agent capability can expand quickly while governance often expands slowly. Adding a new tool, memory store, API, MCP server, CI/CD action, or approval delegation may seem incremental. Collectively, these changes can push autonomy beyond the enterprise's ability to observe, constrain, audit, and recover.

Negative SMI = Agentic Force exceeds Enterprise Gravity

Governance Debt

A negative stability margin should be treated as governance debt: autonomy has been introduced faster than the enterprise control environment can safely contain it.

3. What ASF Is and Is Not

ASF Is	ASF Is Not
An enterprise architecture framework for agentic AI.	A replacement for NIST AI RMF, ISO/IEC 42001, OWASP, or Zero Trust.
An agentic-system-level stability assessment model.	A complete security threat model.
A method for comparing autonomous capability against governance strength.	A regulatory compliance certification.
A practical way to identify governance debt and required control uplift.	A mathematical proof of agent safety.
A complement to AI risk, security, compliance, and management-system frameworks.	A vendor-specific architecture or product pattern.
A decision-support model for autonomy expansion.	A maturity score for the entire enterprise.
An assessment of the agentic system instance: model, runtime, memory, tools, identity, permissions, policies, and operating context.	A standalone model benchmark or isolated prompt evaluation.
Agent-centered, but not agent-only; it requires tool, MCP, memory, identity, and context inventory through the ABOM.	A tool-only or MCP-server-only risk classification.

Important Limitation

ASF should not be used to prove that an agent is safe. It should be used to determine whether the enterprise has enough governance strength to justify the agent's current or proposed level of autonomy.

3.1 Assessment Unit: Agentic System Instance

ASF does not assess an isolated model, standalone prompt, or individual tool in isolation. ASF assesses an agentic system instance: the deployed configuration through which autonomous or semi-autonomous reasoning becomes enterprise action.

An agentic system instance includes the model, instructions and prompts, runtime or orchestrator, memory, tools and MCP servers, APIs and plugins, data sources, identity, permissions, policy controls, approval model, deployment environment, and operating context.

For readability, this paper often uses the term agent. Within ASF, agent should be understood as shorthand for the full agentic system instance.

This distinction matters because Agentic Force does not come from the model alone. Force emerges from the combination of decision autonomy, available tools, accessible context, memory, permissions, delegated authority, and operating environment.

Tools and MCP servers should be governed capabilities in their own right, but their operational impact depends on which agentic system can invoke them, under what authority, in which context, and through which controls.

ASF is therefore agent-centered, but not agent-only. It evaluates the full decision-action system.

Multi-Agent Workflows

Where several agents collaborate to complete a workflow, the assessment unit is the workflow as a whole, including its orchestrator and the handoff paths between agents. Collaboration Complexity exists as an AFI dimension precisely so that multi-agent coordination is scored as a property of one instance rather than distributed across several assessments.

Individual agents within the workflow may be assessed separately as a diagnostic aid, and doing so is often useful for locating where uplift is required. Such sub-assessments are supporting analysis. The deployment decision is made against the workflow assessment, because the blast radius of a multi-agent workflow is not the maximum of its parts.

Where a sub-agent is independently addressable — invocable outside the workflow, holding its own credentials, or reachable by another consumer — it constitutes a separate agentic system instance and requires its own assessment.

3.2 Deployment Assessment and Product Assessment

Section 3.1 defines the assessment unit as a deployed agentic system instance. A software distribution, agent framework, or reusable agentic product is not such an instance. The same product deployed under a shared human credential and under a dedicated workload identity presents different Identity, Authorization, and Recovery characteristics, and may present different Operational Reach and Criticality. Those dimensions are properties of the deployment, not of the artifact.

ASF therefore defines two assessment modes.

Mode	Status	Produces	Applies To
Deployment Assessment	Normative	AFI, EGI, SMI and an Orbit, supporting a deployment decision.	A specific deployed instance in a specific environment.
Product Assessment	Descriptive	A capability envelope. It does not produce an Orbit.	A distribution, framework, or reusable agentic product.

All orbit classifications in this specification refer to Deployment Assessment.

A Product Assessment reports what the product enforces by default, what it enables, what it delegates to the deploying environment, and the range of force it can reach under minimum and maximum configuration.

Assessment Mode Rule

Assigning an Orbit to a product independently of deployment context is a misuse of the framework. Where a product-level score is required for procurement or comparison, it must be reported as a capability envelope with the deployment assumptions stated.

4. Related Work and ASF Differentiation

ASF does not claim to invent autonomy-proportional governance. The field is converging on the idea that AI-agent autonomy should be earned, bounded, evidenced, and matched with proportional oversight. ASF builds on that direction and contributes a practical enterprise-architecture method for applying it: a repeatable way to assess one agentic system at a time, in its specific operating context.

Neighboring Work	Primary Focus	How ASF Differs
OWASP Top 10 for Agentic Applications 2026	Security risks in autonomous and agentic systems that plan, act, and make decisions across workflows.	ASF uses agentic security guidance as input, but focuses on architecture stability: whether governance strength is proportional to agentic force.
Singapore Model AI Governance Framework for Agentic AI	Responsible deployment of agents through bounded powers, accountability, human oversight, lifecycle controls, and technical safeguards.	ASF operationalizes proportional oversight through AFI, EGI, SMI, evidence tiers, deployment gates, and review packets.
Pre-deployment assurance research for enterprise AI agents	Operational envelopes, scenario-based verification, trust certificates, and deployment verdicts such as Approved, Conditional, and Rejected.	ASF complements assurance methods with a governance-capacity model and per-agent, per-context architecture review discipline.
Zero Trust and policy-as-code approaches	Continuous verification, least privilege, policy enforcement, and access decisions.	ASF extends these concepts to agent identity, tool calls, memory, context retrieval, runtime plans, and autonomous actions.

The honest contribution of ASF is therefore narrow but useful: it packages autonomy-proportional governance into a repeatable assessment method that measures two sides of the equation — the operational pressure created by the agent and the governance gravity available to contain it.

ASF should be read as a practical operating model for enterprise architects, not as a replacement for AI governance, agentic security guidance, or formal assurance research.

5. Core Model: AFI, EGI, SMI

ASF evaluates each agentic system instance in a defined operating context. It does not assign one maturity level to the whole enterprise because different agentic systems create different forces and operate under different controls.

The assessment unit is one agentic system instance: one agentic configuration, in one business context, with a defined set of users, tools, MCP servers, permissions, data sources, memory capabilities, runtime controls, and deployment environments.

5.1 Agentic Force Index

AFI measures the operational pressure created by the agentic system as deployed, including its autonomy, tools, memory, permissions, identity delegation, and operating context. It is not inherently good or bad. It is a measurement of force created by the deployed decision-action capability.

AFI Dimension	Measures
Decision Autonomy	How independently the agent can decide and act.
Decision Velocity	How frequently the agent makes or triggers decisions.

AFI Dimension	Measures
Operational Reach	How broadly the agent can affect systems, domains, external parties, and people.
Operational Criticality	How severe the impact could be if the agent acts incorrectly.
Adaptability	Whether the agent can dynamically change plans, strategies, or tools.
Collaboration Complexity	Whether the agent coordinates with humans, other agents, or multi-agent workflows.
Persistence and Memory	How long the agent persists and whether it maintains reusable memory.

ASF deliberately includes Operational Criticality inside the Agentic Force Index because consequence severity raises the required governance threshold directly. An agent that can affect high-criticality systems creates more enterprise pressure than an agent with the same autonomy operating in a low-impact context.

Operational Reach: Systems and People

Operational Reach is scored as the wider of two measures: the systems the agent can reach, and the population it serves. An agent touching one application but serving an entire enterprise has enterprise reach.

Score	System breadth	Consumer population
0	No system access	No consumers
1	Single user workspace	Single user
2	Single application or repository	Single team
3	Multiple apps in one domain	Single business unit or department
4	Multiple domains or business units	Multiple business units
5	Enterprise-wide or external ecosystem	Enterprise-wide, customer-facing, or external parties

Where the two measures differ, the higher score applies and the assessment must record which measure drove it. An agent serving external customers is scored 5 on population regardless of how few systems it touches, because an incorrect action reaches parties outside the enterprise boundary.

This dual measure exists because consequence multiplies with exposure. Two agents reading the same knowledge base — one used by a single analyst, one used by forty thousand employees — present materially different risk while touching identical systems. Operational Criticality does not close this gap, because Criticality measures the severity of one incorrect action rather than the number of parties exposed to it.

5.2 Enterprise Gravity Index

EGI measures the enterprise’s ability to observe, constrain, govern, and recover from the full agentic system instance, including controls over tools, MCP servers, memory, identity, policy, and action execution. EGI is tied to evidence tiers so high scores require implementation proof.

EGI Dimension	Measures
Identity	Whether the agent has distinct, verifiable, attributable identity.
Authorization	Whether permissions are scoped, contextual, and enforced.
Policy Enforcement	Whether rules are technically enforced before actions take effect.

EGI Dimension	Measures
Observability	Whether decision provenance can be reconstructed.
Governance and Lifecycle	Whether agents, tools, models, memory, and policies are managed as assets.
Recovery and Resilience	Whether agent behavior can be stopped, reversed, contained, or recovered.
Assurance and Evaluation	Whether behavior is tested, monitored, red-teamed, and evaluated at runtime.

5.3 Stability Margin Index

$$SMI = EGI - AFI$$

SMI is a normalized architecture indicator representing the margin between measured governance capability and measured agentic force. It is not a probability of failure, compliance score, or complete residual-risk calculation.

SMI expresses proportionality. It does not express whether a prerequisite control is present. The final deployment disposition is the more restrictive of the SMI orbit and any ceiling applied by the non-compensatory rules in Section 13.2.

6. Force and Gravity Metaphor

ASF uses centrifugal force as a metaphor for organizational pressure, not as literal physics. In the metaphor, the enterprise control plane is the center: identity, authorization, policy, audit, approved tools, approved data, recovery mechanisms, and accountable ownership.

As agents gain autonomy, action speed, tool access, memory, and cross-domain reach, they create outward pressure on that control plane. Enterprise Gravity is the set of controls that keeps agent behavior in a stable operating boundary.

The goal is not to eliminate force. The goal is to design enough gravity to sustain useful autonomy.

Metaphor Boundary

The metaphor is intended to make the architecture problem memorable. The ASF scoring model is an enterprise assessment model, not a physics equation.

7. Laws, Axioms, and Vocabulary

7.1 Three Laws of Agentic Stability

1. Autonomy generates force: every increase in agent autonomy generates an increase in organizational force.
2. Stability requires proportional gravity: enterprise stability depends on governance increasing at least as fast as agentic capability.
3. Autonomy must not exceed containment: an agent should not operate beyond the enterprise's ability to observe, constrain, audit, and recover from its behavior.

7.2 Four Axioms

4. Agents are decision systems, not merely applications.
5. Every autonomous action has a blast radius.
6. Governance must become runtime-aware.

7. Stability is agent-specific and context-specific.

7.3 Core Vocabulary

Term	Definition
Agent	Readable shorthand for the full agentic system instance unless explicitly stated otherwise.
Agentic Force	Operational pressure created by an agent's autonomy, velocity, reach, criticality, adaptability, collaboration, and memory.
Enterprise Gravity	The organization's ability to observe, constrain, authorize, audit, govern, recover, and assure agent behavior.
Orbit	The stability classification of an agent based on SMI, subject to the non-compensatory rules in Section 13.2.
Governance Debt	The gap created when Agentic Force exceeds Enterprise Gravity.
Control Uplift	Actions required to increase Enterprise Gravity or reduce Agentic Force.
Agentic System Instance	The deployed configuration through which reasoning becomes action: model, prompts, runtime, memory, tools, MCP servers, data sources, identity, permissions, policies, approval model, and operating context.
Control Responsibility	Whether a control is operated by the assessing organization, an internal platform team, or a third-party vendor. See Section 12.1.
Non-Compensatory Ceiling	The worst orbit permitted by the rules in Section 13.2, applied after the SMI orbit is determined.

8. Assessment Methodology

ASF assessment follows a structured workflow. The output is an agent-level stability profile, not a generic enterprise maturity label.

Step	Assessment Activity	Output
1	Identify the agent and owner	Agent profile
2	Define operating context, tools, data sources, users, consumer population, memory, and environment	Operating context
3	Create the Agent Bill of Materials, including Control Responsibility per gravity dimension	ABOM
4	Score Agentic Force	AFI
5	Score Enterprise Gravity using evidence tiers	EGI
6	Calculate Stability Margin	SMI
7	Apply non-compensatory rules and assign final orbit and deployment gate	Decision recommendation
8	Perform threat modeling where required	Threat model and mitigations
9	Define control uplift or risk acceptance	Governance action plan
10	Set reassessment triggers	Review cadence

Assessment Unit

One agentic system instance: a deployed agent configuration operating in one defined business context, with known tools, MCP servers, data sources, users, permissions, memory, identity, policies, approval model, runtime controls, and deployment environment.

Unknown Capability Rule

Evidence tiers discipline gravity, because a control must be shown to work. No equivalent discipline applies to force. An assessor who cannot determine whether an agent holds production credentials, cross-domain reach, persistent memory, or auto-approved execution must not record those dimensions as low.

Unknown is not zero. Where a material AFI capability cannot be verified, the assessment must either be marked incomplete or scored at the highest reasonably plausible value until the capability is bounded. An assessment carrying unverified AFI dimensions must state which dimensions are unverified and must not be presented as a final disposition.

A material change to the deployment zone — promotion from a sandbox or assisted zone to a governed or autonomous operations zone — is a change of operating context, not a numeric modifier. It requires reassessment of the affected AFI and EGI dimensions rather than an adjustment to the computed score.

9. Basic and Advanced Scoring

9.1 Basic Mode

Basic Mode is intended for workshops, first-pass triage, and early governance conversations. All dimensions are equally weighted.

$$\text{AFI} = \text{Total Force Score} / 35 \times 100$$

$$\text{EGI} = \text{Total Gravity Score} / 35 \times 100$$

$$\text{SMI} = \text{EGI} - \text{AFI}$$

9.2 Advanced Mode

Advanced Mode is intended for production, regulated, high-criticality, or high-autonomy agents. It uses domain-specific weighting.

$$\text{Weighted AFI} = \frac{\sum(\text{Force Score} \times \text{Force Weight})}{\sum(\text{Max Score} \times \text{Force Weight})} \times 100$$

$$\text{Weighted EGI} = \frac{\sum(\text{Gravity Score} \times \text{Gravity Weight})}{\sum(\text{Max Score} \times \text{Gravity Weight})} \times 100$$

Agent Type	Higher AFI Weights	Higher EGI Weights
Deployment Agent	Criticality, Reach, Autonomy	Policy, Recovery, Observability
Customer Service Agent	Reach, Criticality, Memory	Authorization, Assurance, Observability
Security Agent	Autonomy, Criticality, Velocity	Identity, Policy, Recovery
Financial Agent	Criticality, Autonomy, Reach	Authorization, Assurance, Governance
Multi-Agent SDLC Agent	Collaboration, Adaptability, Reach	Observability, Policy, Governance

10. Evidence Tiers

ASF requires EGI scores to be evidence-based. A control may exist on paper but fail during runtime. Evidence tiers help distinguish intent from operating effectiveness.



EGI scores above 3 require Tier 3+ evidence. Production-impacting agents require Tier 4 for key controls.

Figure 2. ASF evidence tiers.

Tier	Meaning	Typical Evidence
0	Claimed, no evidence	Verbal statement, slide, or unvalidated design claim.
1	Documented design	Architecture document, policy draft, design decision record.
2	Implemented control	Configured identity, gateway, policy, logging, or workflow.
3	Tested control	Test result, drill, validation evidence, policy test, rollback test.
4	Runtime-verified control	Production trace, policy decision log, runtime monitor, audit event.
5	Continuously monitored and audited control	Dashboards, recurring evidence, automated control checks, periodic audit.

Evidence Rule

No EGI dimension may receive a score above 3 unless the control has Tier 3 or higher evidence.

An agentic system instance with production action authority must hold Tier 4 evidence for Policy Enforcement, Observability, Recovery, and Assurance before it may be classified better than Edge Orbit. Where this condition is not met, Edge Orbit is the maximum permitted classification regardless of SMI, and the assessment must record which dimensions fell short and at what tier.

This is a ceiling on classification, not an adjustment to EGI. The computed indices remain as scored so that the evidence gap is visible rather than absorbed.

11. Threat-Modeling Gate

ASF is not a complete security threat model. High-force agents require threat modeling before production deployment.

Any agent operating in Edge, Unstable, or Critical Orbit must undergo threat modeling. Threat modeling is also required when the agent has production action authority, sensitive data access, multi-agent handoffs, long-term memory, or external-facing exposure. A positive SMI does not waive these trigger-based requirements.

Threat Category	Assessment Questions
Prompt injection	Can user or retrieved content override the agent's instructions or policy boundaries?
Tool misuse	Can the agent invoke tools in unintended order, scope, or context?
Excessive agency	Does the agent have broader permissions than its task requires?
Identity abuse	Can agent identity or delegated user authority be misused?
Memory/context poisoning	Can malicious content persist and influence future behavior?
Data exfiltration	Can retrieved or generated content leak sensitive data?
Inter-agent manipulation	Can one agent manipulate another through context, task handoff, or shared memory?
Policy evasion	Can the agent bypass gateway, broker, or policy enforcement points?
Cascading failures	Can one agent action trigger uncontrolled downstream effects?
Recovery failure	Can the enterprise stop, reverse, or contain the action fast enough?

12. Agent Bill of Materials

Every ASF assessment should include an Agent Bill of Materials (ABOM). The ABOM defines the assessment boundary for the agentic system instance. It is the inventory of components and dependencies required to understand what the system is, what it can access, which tools or MCP servers it can invoke, and what it can affect.

ABOM Category	Required Information
Agent	Name, version, owner, purpose, business domain.
Model	Provider, model, deployment mode, approved model status.
Runtime	Agent framework, orchestrator, hosting environment.
Prompts	System prompts, task prompts, prompt versioning and approval.
Tools	APIs, MCP servers, plugins, scripts, functions, workflows; read/write capability, tool risk level, and action reversibility.
Data Sources	Knowledge bases, vector stores, databases, files, SaaS systems.
Memory	Session, short-term, long-term, enterprise memory.
Identity	Agent identity, user delegation, service identity.
Authorization	Permissions, scopes, roles, environments, JIT access, delegated authority, and on-behalf-of model.
Policies	Policy rules, enforcement points, exceptions, tool/action policies, and policy decision logs.
Observability	Logs, traces, audit stores, dashboards, and prompt-plan-tool-action traceability.
Recovery	Rollback, kill switch, circuit breakers, compensating actions, and reversibility limits.
Assurance	Evaluations, red-team tests, runtime monitors, drift checks.
Dependencies	External services, gateways, brokers, registries, MCP servers, and connector dependencies.
Consumer Population	Who the agent serves and how many: single user, team, business unit, multiple business units, enterprise-wide, or external parties.
System Boundary	Scope of the assessed agentic system instance, including runtime, tools, memory, identity, and deployment context.
Tool / MCP Classification	For each tool or MCP server: data sensitivity, action impact, reversibility, auth model, auditability, and abuse potential.
Action Authority	Whether actions are read-only, draft-only, human-approved, policy-approved, or independently executed.
Control Responsibility	For each EGI dimension: self-operated, platform-operated, or vendor-operated. See Section 12.1.

ABOM Rule

No agentic system should move beyond Controlled Orbit without an Agent Bill of Materials that defines the system boundary and classifies tools, MCP servers, memory, identity, permissions, and action authority.

12.1 Control Responsibility

An assessing team does not always own the controls it is scoring. A team consuming a vendor agent platform, a low-code automation product, or a managed agent service may have no ability to change Identity, Authorization, Policy Enforcement, or Recovery, because those are properties of the platform. The same team typically retains Governance and Lifecycle, retains Observability of its own workloads, and can partially compensate on Authorization through the credentials and scopes it issues to the platform.

For each EGI dimension, the ABOM records who operates the control.

Value	Meaning
Self-operated	The assessing organization implements and can change the control.
Platform-operated	An internal platform team operates the control on the assessing team's behalf.
Vendor-operated	A third party operates the control. The assessing organization can influence it only contractually.

Control Responsibility Does Not Affect Scoring

A weak authorization boundary presents the same risk whether the assessing team wrote it or bought it. Scoring an agent more favourably because its controls are outside the team's reach would reward the absence of remediation capability, and would conflict with Section 18.1, which warns against accepting vendor self-certification in place of enterprise validation. Control Responsibility does not affect AFI, EGI, SMI, or Orbit.

Control Responsibility governs remediation routing. Where a dimension requiring uplift is vendor-operated, the uplift action is not to implement the control directly. It is one of the following:

- Request the capability from the vendor and record the commitment and date.
- Obtain contractual or attestation evidence sufficient to raise the evidence tier.
- Compensate outside the platform — for example enforcing policy at an egress gateway the organization does control.
- Reduce force instead of raising gravity, by narrowing reach, criticality, or autonomy until the margin is acceptable.
- Accept the risk through the Section 13.3 workflow, with expiry.

Where every uplift path for a mandatory control is vendor-operated and no compensating control exists, the correct outcome is a reduction in force or a formal risk acceptance — not a more favourable score.

Control Responsibility is also the field that connects the two assessment modes defined in Section 3.2. A vendor's own agent product is assessed under Product Assessment, which produces a capability envelope and no Orbit. A consuming organization assesses its deployment of that product under Deployment Assessment. Control Responsibility records which dimensions of the consumer's Deployment Assessment are determined by the vendor's Product Assessment.

13. Orbit Classification and Deployment Gates

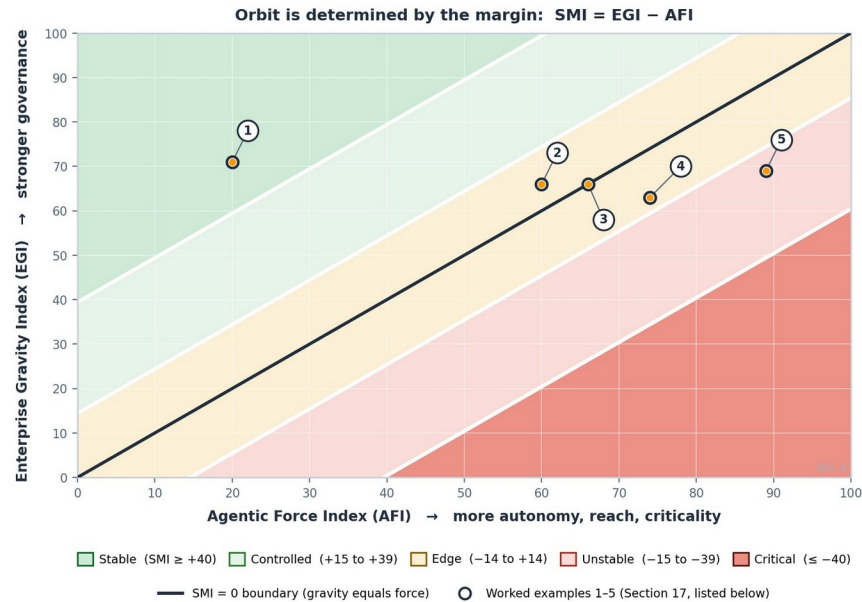


Figure 3. Orbit classification by AFI and EGI.

The numbered points on the chart correspond to the five worked examples introduced in Section 17.

#	Worked Example	AFI → EGI → SMI	Orbit
1	Documentation Assistant	20 → 71 → +51	Stable
2	Client-Request Agent	60 → 66 → +6	Edge
3	K8s Deployment Agent	66 → 66 → 0	Edge
4	Multi-Agent SDLC	74 → 63 → -11	Edge
5	Autonomous Remediation	89 → 69 → -20	Unstable

13.1 Orbit Bands

SMI	Orbit	Default Deployment Decision
$SMI \geq +40$	Stable Orbit	Approved for scale with monitoring.
$+15 \leq SMI < +40$	Controlled Orbit	Approved with standard controls.
$-14 \leq SMI < +15$	Edge Orbit	Pilot, sandbox, or limited production only. Active control management required.
$-39 \leq SMI < -14$	Unstable Orbit	Do not expand. Control uplift required.
$SMI < -39$	Critical Orbit	Do not deploy without formal exception.

Basic Mode SMI occurs in increments of approximately 2.857, so the band boundaries are stated as inequalities rather than integer ranges. The classification is identical to the behaviour of the ASF Assessment Workbook.

13.2 Non-Compensatory Rules

AFI and EGI are compensatory indices: a high score in one dimension is offset by low scores in others. This is appropriate for measuring overall force and overall gravity, but it permits two failure modes that a margin alone cannot express.

First, an agent may combine extreme autonomy with extreme criticality while scoring low on unrelated dimensions, producing a moderate AFI and a favourable orbit. Second, an agent may be missing a prerequisite control entirely while compensating through strength in five other governance dimensions, producing a favourable EGI.

Some conditions are not matters of degree. ASF applies non-compensatory rules after the SMI orbit is determined. Each rule may only lower a classification; none may raise one.

Rule 1 — Force Concentration

Where Decision Autonomy is scored 4 or higher and Operational Criticality is scored 4 or higher, the agentic system instance must not be classified better than Edge Orbit, regardless of SMI.

The rule is conjunctive by design. Autonomy without blast radius describes an agent acting freely inside a sandbox. Blast radius without autonomy describes a consequential agent under human approval on every action. Neither alone warrants a ceiling. Their combination — an agent that both acts independently and can cause severe harm — is the condition that a margin cannot safely offset.

Rule 2 — Mandatory Control Floor

For an agentic system instance with production action authority and Decision Autonomy scored 3 or higher, where Identity, Authorization, or Policy Enforcement is scored at or below 1, a ceiling applies.

Lowest mandatory control score	Maximum permitted orbit
0 — control absent	Unstable Orbit
1 — control nominal	Edge Orbit
2 or above	No ceiling applied

Identity, Authorization, and Policy Enforcement are treated as prerequisites rather than contributors. Observability, Recovery, and Assurance describe how well an agent's behaviour can be seen, reversed, and validated; they cannot substitute for the absence of an identity boundary or an enforcement point.

Rule 3 — Evidence Gate

An agentic system instance with production action authority that does not meet the Tier 4 evidence requirement stated in Section 10 must not be classified better than Edge Orbit. This rule formalizes an existing requirement rather than introducing a new one.

Final Disposition

The deployment decision is the more restrictive of the SMI orbit and any ceiling applied by Rules 1, 2, and 3. Where a ceiling applies, the assessment must record which rule fired and on which dimension.

13.3 Risk Acceptance Workflow

If SMI is negative, the enterprise must choose one of five actions: reduce force, increase gravity, restrict deployment zone, add approval, or formally accept risk with expiration.

Risk Acceptance Field	Requirement
Business Owner	Named accountable owner.
Technology Owner	Named accountable owner.
Reason	Why exception is needed.
Compensating Controls	Controls that reduce exposure.
Expiration Date	No open-ended exceptions.
Reassessment Date	Required follow-up.
Residual Risk	Accepted risk statement.
Approval Record	Governance approval evidence.

A ceiling applied under Section 13.2 cannot be resolved by risk acceptance alone. Risk acceptance records that the enterprise has chosen to operate with a known gap; it does not raise the classification. The underlying dimension must move, or force must be reduced, for the ceiling to lift.

Where a mandatory control is vendor-operated and no compensating control exists, risk acceptance or force reduction is the correct outcome, as described in Section 12.1.

14. Reference Architecture

The ASF reference architecture defines how enterprises create enough Enterprise Gravity to safely contain Agentic Force. The architecture is vendor-neutral and focuses on control responsibilities rather than product selection. Figure 4 is a simplified view; the table below enumerates the full set of control layers.

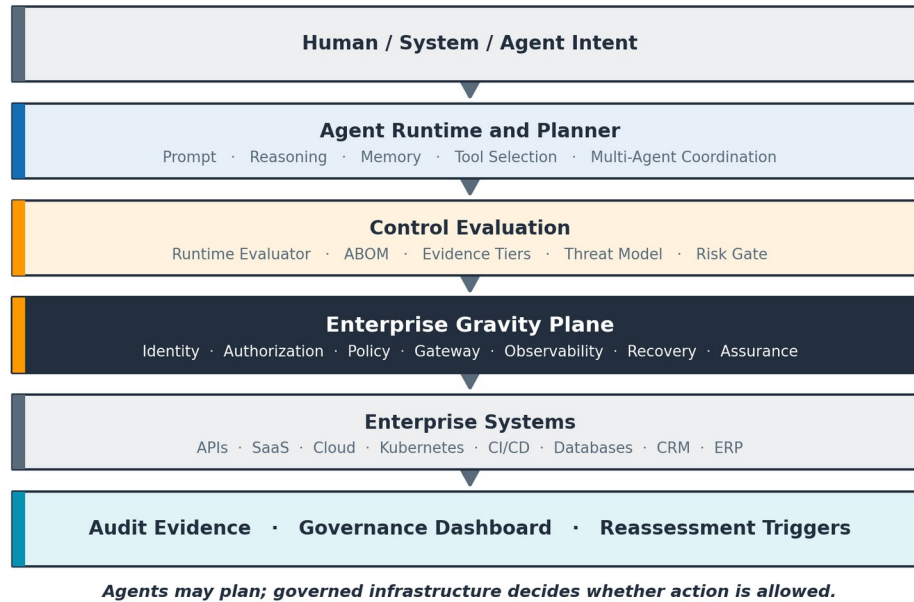


Figure 4. ASF reference architecture.

Layer	Purpose	Representative Controls
Agent experience	Receives intent from humans, systems, or agents.	Request identity, authority, intent classification.
Agent runtime and planner	Generates plans and selects tools.	Agent identity, prompt governance, plan traceability.
Context and memory	Provides enterprise knowledge and persistent state.	Data classification, retrieval authorization, memory governance.
Tool registry and broker	Controls what tools the agent can invoke.	Tool inventory, schema validation, least privilege, action routing.
Policy and gateway	Enforces enterprise rules before action.	PDP/PEP, OPA/Cedar-style policy, API/AI gateway, allowlists.
Runtime evaluation	Evaluates plans, inputs, outputs, and tool behavior.	Prompt-injection checks, drift detection, business-rule validation.
Observability and audit	Reconstructs decisions and outcomes.	Prompt-plan-tool-action traces, SIEM, audit evidence.
Recovery and containment	Stops or reverses unsafe behavior.	Kill switch, rollback, circuit breaker, credential revocation.
Governance dashboard	Maintains portfolio visibility.	Agent catalog, AFI/EGI/SMI, evidence tiers, review dates.

Architecture Rule

Agents may plan, but governed infrastructure decides whether action is allowed.

14.1 Tool and MCP Governance Boundary

A tool or MCP server should be classified for data sensitivity, action impact, reversibility, authorization model, auditability, and abuse potential. However, tool classification alone does not determine stability.

Stability depends on how that tool is exposed to a specific agentic system, under a specific identity, with specific permissions, policies, memory, and approval controls. The same MCP server may be low-force in a read-only assistant and high-force in an autonomous approval or deployment workflow.

15. Patterns and Anti-Patterns

15.1 Recommended Patterns

Pattern	Use When	Minimum Controls
Read-Only Assistant	Agent summarizes or answers from approved sources.	User identity, source authorization, basic logging.
Human-Approved Action	Agent prepares action but human approves.	Agent registry, approval workflow, action preview, tool-call logs.
Policy-Approved Action	Agent executes under strict policy.	Workload identity, broker, runtime policy, rollback.
Multi-Agent Workflow	Several agents coordinate across process.	Agent-to-agent identity, handoff tracing, shared-context governance.
Autonomous Operations	High-autonomy production workflows.	Dynamic authorization, continuous assurance, blast-radius isolation, formal risk governance.

15.2 Anti-Patterns

Anti-Pattern	Why It Is Dangerous
Direct-to-System Agent	Bypasses tool broker, policy, approval, and audit controls.
Shared Credential Agent	Breaks attribution and makes revocation difficult.
Invisible Agent	Exists outside catalog, ownership, reassessment, and risk visibility.
Prompt-Only Governance	Relies on instructions rather than enforceable controls.
Human Rubber Stamp	Creates approval theater without meaningful accountability.
Ungoverned Tool Discovery	Lets the agent invoke unapproved tools and dependencies.
Product Orbit Assignment	Assigns a stability classification to a distribution independently of deployment context, where identity, authorization, and recovery are undetermined.

16. Domain Overlays

The base ASF model is generic. Domain overlays adjust weights, evidence expectations, threat scenarios, and control uplift recommendations for specific types of agents.

Overlay	When to Use	Required Emphasis
Software Delivery Agents	Agents reviewing, coding, testing, merging, or deploying software.	Artifact provenance, repo permissions, CI/CD policy, rollback, code review.
Cloud and Infrastructure Agents	Agents provisioning, configuring, or remediating cloud/platform systems.	Least privilege, blast-radius isolation, IaC policy, recovery drills.
Security Operations Agents	Agents triaging alerts, changing controls, or responding to incidents.	Identity, privilege boundaries, auditability, containment.
API Governance Agents	Agents reviewing specs, attaching policies, publishing APIs.	Catalog workflow, policy binding approval, traceability.
Customer Service Agents	Agents accessing CRM, cases, customer data, or external communications.	PII controls, authorization, assurance, human escalation.
Financial Operations Agents	Agents approving payments, invoices, reconciliations, or financial actions.	Segregation of duties, approval evidence, fraud controls.
HR and Workforce Agents	Agents affecting employees, access, payroll, or performance processes.	Sensitive data controls, human accountability, audit.
Regulated-Data Agents	Agents touching health, financial, legal, or regulated datasets.	Data minimization, audit-grade evidence, regulatory review.
Multi-Agent Engineering Workflows	Planner, coder, tester, reviewer, releaser collaboration.	Handoff tracing, role boundaries, conflict resolution, shared memory governance.

Status of Advanced Mode Weights

The domain overlay weights published in the ASF Assessment Workbook are reference calibration profiles reflecting expert judgment. They are not empirically validated coefficients, and no claim is made that they predict incident likelihood.

The workbook is the normative source for these values. Organizations may substitute their own weights with documented governance approval, and must state which profile was used when reporting an Advanced Mode result. Two assessments are comparable only where the same overlay, the same weights, and the same scoring mode were applied.

17. Worked Examples

The five examples below are scored in Basic Mode. Their classifications are unchanged from ASF v1.0; the non-compensatory rules introduced in Section 13.2 do not alter any of them.

Agent	Purpose	AFI	EGI	SMI	Orbit	Decision	Required Uplift
Documentation Assistant	Knowledge answers	20	71	+51	Stable	Scale	Basic source auth and logging.
Client-Request Automation Agent	Client-servicing actions	60	66	+6	Edge	Pilot / limited production	Runtime eval; audit trace; rollback evidence; advisor gates.

Agent	Purpose	AFI	EGI	SMI	Orbit	Decision	Required Uplift
Kubernetes Deployment Agent	Deploy workloads	66	66	0	Edge	Limited production	Runtime assurance; namespace auth; rollback evidence.
Multi-Agent SDLC Workflow	Plan/code/test/release handoffs	74	63	-11	Edge	Pilot only	Threat model; handoff tracing; shared-context governance.
Autonomous Remediation Agent	Incident remediation	89	69	-20	Unstable	Do not expand	Tier 4+ evidence; blast-radius isolation; formal exception or uplift.

17.1 Anonymized Client-Request Automation Agent (Financial Services)

This example is drawn from a representative financial-services client-servicing pattern; client, vendor, and identifying architecture details have been generalized. It exercises the full framework and illustrates ASF's central insight: strong enterprise governance does not guarantee a positive stability margin when agentic force is high. The scores below are representative; a live assessment would attach real evidence at the tiers shown.

Agent and operating context: the agent receives client-servicing requests — for example contact-detail updates, document requests, and transfers between a client's own accounts — interprets intent, retrieves client context, and prepares or executes the servicing action through backend APIs. Higher-impact actions require human advisor approval; lower-impact actions execute under policy. The agent acts under delegated, on-behalf-of authority and is fronted by an API gateway, enterprise identity provider, tool broker, and runtime inspection policy at defined control points.

ABOM Category	Anonymized Contents
Agent purpose	Interpret client-servicing requests, retrieve client context, prepare or execute servicing actions, and route high-impact requests for advisor approval.
Runtime	Enterprise agent runtime integrated with a bounded servicing-action catalog, tool broker, API gateway, and client-servicing workflow.
Tools	Servicing APIs with write access, CRM read access, document service, policy/risk classifier, advisor approval workflow, and notification channel.
Data sources	Client CRM and account systems containing regulated PII and financial data; approved servicing policy and procedure sources.
Memory	Session and task memory only; no long-term, cross-client, or self-maintained enterprise memory.
Identity and authorization	Distinct agent identity with on-behalf-of user delegation; tool and environment scopes enforced through gateway and broker controls.
Consumer population	Client-servicing staff within one business unit, acting on behalf of external clients.
Control Responsibility	Identity, Authorization, and Policy Enforcement self-operated; Observability platform-operated; Recovery and Assurance self-operated.
Explicit restrictions	No autonomous high-impact financial action, no cross-client memory, no self-approved exceptions, and no bypass of advisor approval or policy gates.

AFI Dimension	Score	Rationale
Decision Autonomy	3	Executes low-impact actions under policy; high-impact financial or client-impacting actions require explicit advisor approval.

AFI Dimension	Score	Rationale
Decision Velocity	3	Event-driven decisions triggered by client-servicing requests.
Operational Reach	3	Touches multiple applications within one business domain, including servicing, CRM, document, and account-related APIs. Consumer population is one business unit, which does not raise the score.
Operational Criticality	5	Incorrect action can affect regulated client financial records, account servicing, compliance exposure, and client trust.
Adaptability	3	Generates plans dynamically over retrieved client context, but remains bounded by a defined servicing-action catalog.
Collaboration Complexity	2	Collaborates with advisors and workflow systems; no dynamic multi-agent coordination.
Persistence and Memory	2	Uses short-term task memory only; no long-term or cross-client memory.
Total	21 / 35	AFI = 60

EGI Dimension	Score	Tier	Evidence Example
Identity	4	3	Distinct agent identity with strong authentication and on-behalf-of delegation; tested evidence required to sustain a 4.
Authorization	4	3	Least privilege by tool and environment, gateway-enforced, with delegated context carried through the request path.
Policy Enforcement	4	4	Runtime enforcement at the gateway with external inspection at defined control points; runtime-verified.
Observability	3	3	Gateway and tool-call logs plus partial prompt, plan, and action traces; not yet full audit-grade end-to-end provenance.
Governance and Lifecycle	3	3	Catalogued with lifecycle approval and reviewed by AI risk, architecture, and change governance.
Recovery and Resilience	3	3	Gateway kill switch and advisor approval gates exist; rollback or compensating control for executed financial actions is only partly tested.
Assurance and Evaluation	2	2	Pre-release scenario testing exists; no continuous runtime evaluation, drift detection, or outcome validation yet.
Total	23 / 35	—	EGI = 66

$$\text{SMI} = 66 - 60 = +6 \quad \text{Orbit: Edge}$$

Non-compensatory check: Decision Autonomy is 3, so Rule 1 does not apply. Identity, Authorization, and Policy Enforcement are all above 1, so Rule 2 does not apply. The agent carries production action authority and does not hold Tier 4 evidence for Observability, Recovery, and Assurance, so Rule 3 imposes an Edge ceiling. The SMI orbit is already Edge, so the final disposition is unchanged — but the assessment records the evidence gap explicitly.

Interpretation: the agent is useful and meaningfully governed, but it is still close to the governance boundary because it can touch regulated client data and financial-servicing actions. It can operate in pilot or limited production, but it should not

receive broader autonomy until runtime assurance, audit-grade decision provenance, and tested recovery controls improve.

Control Uplift	Reason
Add runtime evaluation for client-servicing decisions	High-impact or ambiguous client requests should be evaluated at runtime before execution or approval routing.
Increase observability to audit-grade decision provenance	Reviewers should reconstruct the initiating user, delegated authority, prompt, retrieved client context, plan, policy result, tool call, approval decision, action, and outcome end to end.
Test rollback and compensating controls	Some financial or client-record actions cannot be simply undone; compensating processes must be tested and evidenced.
Keep advisor approval for high-impact workflows	Human accountability remains necessary for transfers, sensitive data changes, exceptions, and elevated-risk servicing actions until SMI improves.
Require threat modeling before autonomy expansion	The agent touches regulated PII and financial workflows; model prompt injection, tool misuse, delegated-authority abuse, approval bypass, and data-exfiltration paths.

Deployment decision: pilot or limited production only. Keep advisor approval for high-impact actions, retain policy-based controls for lower-impact actions, and do not expand autonomy until the identified control uplifts are implemented and reassessed.

18. Misuse, Limitations, and Threats to Validity

18.1 Misuse Guardrails

- Do not use SMI as the only approval decision.
- Do not claim that a positive SMI proves an agent is safe.
- Do not replace threat modeling, security testing, regulatory review, or business risk assessment with ASF.
- Do not average all agents into a single enterprise maturity score.
- Do not assign high EGI scores based only on design intent.
- Do not treat human approval as sufficient governance by itself.
- Do not let vendors self-certify agent stability without enterprise validation.
- Do not assign an Orbit to a product or distribution independently of deployment context.
- Do not score a control more favourably because the assessing team cannot change it.
- Do not ignore domain-specific risk overlays.

18.2 Limitations and Threats to Validity

ASF is a decision-support framework, not a mathematical proof of agent safety. Its results should guide architecture, governance, and risk discussions, but they should not be treated as automatic approval decisions.

Limitation	Why It Matters	Mitigation
Scoring is judgment-based	Two qualified assessors may score the same agent differently, especially when evidence is incomplete.	Use calibration sessions, scoring examples, review boards, and evidence-tier requirements.
Inter-rater reliability is unmeasured	One rubric point equals approximately 2.857 SMI points while orbit bands are 25 to 29 points wide, so modest disagreement between assessors can change a classification.	An inter-rater reliability study across multiple assessors and agents is required before thresholds are treated as calibrated.
Orbit thresholds are heuristic	The +40, +15, -14, and -39 thresholds are pragmatic starting points, not empirically derived safety boundaries.	Calibrate thresholds over time using incident history, regulatory exposure, risk appetite, and domain experience.
Non-compensatory thresholds are reference values	The Rule 1 and Rule 2 trigger levels are informed by analysis of the scoring model, not by incident data.	Review the thresholds against organizational incident history and adjust with documented governance approval.
Advanced-mode weights are contextual	Domain overlays reflect expert judgment and may differ by organization or industry.	Document weights, review them periodically, and require governance approval for high-impact overlays.
Positive SMI does not prove safety	A positive margin indicates proportional governance under the model, not guaranteed safe behavior.	Combine ASF with threat modeling, security testing, runtime monitoring, and regulatory review.
ASF depends on evidence quality	Weak or untested evidence can create false confidence in Enterprise Gravity.	Require tested or runtime-verified evidence for high EGI scores and production-impacting agents.
Control Responsibility is descriptive	It records who operates a control but deliberately does not affect the score, so a	Route remediation through Section 12.1 and use force reduction or risk acceptance where

Limitation	Why It Matters	Mitigation
	consuming organization carries the same classification as the operator of the weak control.	no uplift path exists.
ASF does not replace domain expertise	Healthcare, financial, HR, safety, and regulated workflows require specialized review.	Use ASF alongside domain-specific risk, legal, compliance, clinical, or security disciplines.

19. Standards Alignment

Framework	Primary Purpose	ASF Relationship
NIST AI RMF	AI risk management across lifecycle through Govern, Map, Measure, and Manage.	ASF provides agent-level stability scoring inside broader AI risk management.
ISO/IEC 42001	Requirements and guidance for an AI management system.	ASF provides an architecture assessment method for agentic systems under an AIMS.
OWASP GenAI / Agentic AI	Security threats and mitigations for LLM and agentic systems.	ASF uses OWASP guidance as input to Assurance, Policy, and threat-model gates.
OWASP Multi-Agent Threat Modeling	Threat modeling guidance for systems with multiple coordinating autonomous agents.	ASF requires threat modeling for multi-agent workflows and for Edge, Unstable, or Critical Orbit agents.
Zero Trust	Continuous verification of access to resources.	ASF extends verification to agent identity, delegated authority, tool calls, context, memory, and actions.
OPA / Cedar-style policy	Externalized policy and authorization decisions.	ASF treats policy engines as Enterprise Gravity components.

20. Implementation Roadmap

Phase	Objective	Deliverables
1. Discover	Identify and register active and planned agents.	Agent catalog, owner list, initial ABOMs.
2. Baseline	Run Basic Mode scoring for top agents.	AFI, EGI, SMI, orbit distribution.
3. Govern	Create deployment gates, evidence tiers, and threat-model triggers.	Architecture review packet, risk acceptance workflow.
4. Implement Gravity	Deploy identity, authorization, policy, gateway, observability, recovery, assurance controls.	Reference control plane, dashboard, evidence automation.
5. Automate	Embed ASF into onboarding and CI/CD or agent platform workflows.	Automated metadata validation, policy checks, reassessment triggers.
6. Improve	Use lessons from incidents, red teams, and audits to refine overlays.	Domain overlays, benchmark library, external review.

20.1 Evaluated and Deferred

The following changes were evaluated during the preparation of v1.1 and deliberately deferred. They are recorded so that their absence reads as a decision rather than an oversight.

Deferred Change	Reason
Per-dimension evidence tiers	Adds seven assessment inputs and requires seven evidence artifacts to be sourced per assessment. The precision is diagnostically valuable and is being evaluated as an uplift-planning aid rather than a scoring input.
Effective Gravity = $\min(\text{capability}, \text{evidence tier})$	Makes the requirement that gravity be evidenced mathematically true, but under realistic evidence distributions it renders Stable Orbit unreachable and makes Unstable

Deferred Change	Reason
	the modal classification. It cannot be adopted without simultaneous band recalibration.
Orbit band recalibration	Bands must be wider than rater disagreement and narrower than material change in an agent. Establishing the lower bound requires an inter-rater reliability study.
Graded force ceilings	A graded ceiling would rank concentrated-force and diffuse-force agents in the correct order, where Rule 1 leaves them at the same classification. It would also reclassify a published worked example. Removing the inversion is the defensible claim; strict ordering is not worth the reclassification.
Deployment zone as a numeric modifier	A zone change is a material change of operating context requiring reassessment of the affected dimensions, not a term in the formula.

21. Conclusion

Agentic AI changes the enterprise architecture problem because agents do not merely generate content; they can take action. Enterprises need a way to determine when autonomy is proportional to control strength.

ASF provides that discipline through Agentic Force, Enterprise Gravity, Stability Margin, Orbit Classification, evidence tiers, threat-model gates, ABOM, and control uplift planning. Version 1.1 adds the recognition that proportionality alone is not sufficient: some controls are prerequisites rather than contributors, and some combinations of force cannot be offset by a favourable margin.

The goal is not to slow down agentic adoption. The goal is to make autonomy safe enough to scale.

Increase autonomy only when gravity is strong enough to hold it.

22. Change History

ASF v1.1 retains the v1.0 core model without modification. AFI, EGI, and SMI are computed identically, the seven force dimensions and seven gravity dimensions are unchanged, and no new assessment input is required. All five worked examples published in v1.0 retain their original classification.

22.1 Scoring Changes

Two rules can change a classification. Both are ceilings; neither can improve one.

Change	Section	Effect
Rule 1 — Force Concentration	13.2	Decision Autonomy ≥ 4 and Operational Criticality ≥ 4 caps the classification at Edge Orbit.
Rule 2 — Mandatory Control Floor	13.2	Identity, Authorization, or Policy Enforcement at 0 caps at Unstable Orbit; at 1 caps at Edge Orbit, for production agents with Decision Autonomy ≥ 3 .
Rule 3 — Evidence Gate	13.2	Formalizes the Tier 4 evidence requirement already stated in v1.0 Section 10, which was previously advisory. Caps at Edge Orbit.

22.2 Clarifications

The following additions carry no computation and require no re-scoring.

Change	Section	Summary
Deployment and Product Assessment modes	3.2	A distribution is not a deployed instance. Product Assessment produces a capability envelope and no Orbit.
Multi-agent boundary	3.1	The assessment unit is the workflow including orchestrator and handoff paths. Independently addressable sub-agents require their own assessment.
Operational Reach includes consumer population	5.1	Reach is scored as the wider of system breadth and the population served. Previously the anchors described systems only.
Unknown Capability Rule	8	Unverified AFI capability must be marked incomplete or scored at the highest plausible value, not recorded as low.
Control Responsibility	12.1	The ABOM records whether each gravity control is self-, platform-, or vendor-operated. Routes remediation; does not affect scoring.
Orbit boundary notation	13.1	Band boundaries stated as inequalities rather than integer ranges, matching the workbook's computed behaviour. No classification changes.
Status of Advanced Mode weights	16	Workbook weights are reference calibration profiles, not validated coefficients. The workbook is the normative source.

22.3 Assessment Workbook Corrections

The following defects in the v1.0 ASF Assessment Workbook are corrected in the v1.1 release. They affected the instrument only; the specification was not in error.

Severity	Defect
Critical	Post-2007 spreadsheet functions were stored without the prefix Microsoft Excel requires, so the orbit

Severity	Defect
	classification and several derived columns returned a name error in Excel. The workbook now uses only functions available in Excel 2007 and later.
Critical	Dashboard aggregates read only the first five assessment rows while row formulas were live to row 104. A sixth agent was excluded from every average and every risk counter — a failure that concealed governance debt rather than exposing it.
Major	The weakest-gravity calculation returned only the first dimension holding the minimum score, so where dimensions tied, the uplift recommendation was determined by column order.
Minor	Blank assessment rows displayed an evidence warning status of OK.
Minor	The repository quick-start instructions referenced a workbook filename that did not exist.

A regression test sheet is included in the v1.1 workbook. It verifies that a newly added assessment row propagates to every aggregate, orbit count, and risk counter.

22.4 Migration

No re-scoring is required. Existing assessments carry forward unchanged, and the new workbook columns compute from data already present. Review any assessment where a non-compensatory rule is recorded as fired: the rule identifies which dimension caused the ceiling and therefore where uplift is required.

Appendix A. AFI Rubric

Decision Autonomy

Score	Description
0	No decision capability
1	Suggests actions only
2	Recommends actions for human approval
3	Executes after explicit human approval
4	Executes under policy-based approval
5	Executes independently within delegated authority

Decision Velocity

Score	Description
0	No decisions
1	Occasional or manually triggered
2	Daily or batch decisions
3	Hourly or event-driven
4	Near-real-time
5	Continuous autonomous loop

Operational Reach

Score the wider of system breadth and consumer population. Record which measure drove the score.

Score	System breadth	Consumer population
0	No system access	No consumers
1	Single user workspace	Single user
2	Single application or repository	Single team
3	Multiple apps in one domain	Single business unit or department
4	Multiple domains or business units	Multiple business units
5	Enterprise-wide or external ecosystem	Enterprise-wide, customer-facing, or external parties

Operational Criticality

Score	Description
0	No operational impact
1	Low impact

Score	Description
2	Team productivity impact
3	Application or workflow impact
4	Production, customer, or security impact
5	Financial, regulatory, safety, or enterprise-critical impact

Adaptability

Score	Description
0	No adaptation
1	Static prompt or fixed workflow
2	Chooses predefined paths
3	Generates plans dynamically
4	Learns from feedback or memory
5	Discovers tools, modifies strategies, self-optimizes

Collaboration Complexity

Score	Description
0	No collaboration
1	Single-agent interaction
2	Human-agent collaboration
3	Multiple agents with fixed roles
4	Dynamic handoffs
5	Swarm-like or emergent behavior

Persistence and Memory

Score	Description
0	No persistence
1	Single-session only
2	Short-term task memory
3	Long-lived scoped memory
4	Cross-session memory
5	Enterprise memory or self-maintained knowledge state

Appendix B. EGI Rubric

Score only controls that are present, well-designed, and evidenced at the required tier. Identity, Authorization, and Policy Enforcement are subject to the mandatory control floor in Section 13.2.

Identity

Score	Description
0	No distinct identity
1	Shared user or service account
2	Dedicated service identity
3	Workload identity with ownership metadata
4	Short-lived credentials and strong authentication
5	Cryptographic workload identity with continuous verification

Authorization

Score	Description
0	No authorization boundary
1	Broad static access
2	Role-based access
3	Least privilege by tool or workflow
4	Context-aware authorization
5	Dynamic policy-driven JIT access

Policy Enforcement

Score	Description
0	No policy enforcement
1	Human guidelines only
2	Static configuration checks
3	CI/CD or pre-execution policy checks
4	Runtime policy enforcement
5	Adaptive policy enforcement with risk-based decisions

Observability

Score	Description
0	No logs

Score	Description
1	Basic application logs
2	Tool-call logs
3	Prompt, plan, action, and response traces
4	End-to-end traceability
5	Full decision provenance with audit-grade evidence

Governance and Lifecycle

Score	Description
0	No inventory
1	Informal ownership
2	Agent catalog or registry
3	Lifecycle approval workflow
4	Versioning, certification, and deprecation controls
5	Enterprise control plane for agents, tools, models, memory, and policies

Recovery and Resilience

Score	Description
0	No recovery mechanism
1	Manual intervention only
2	Basic rollback
3	Kill switch or circuit breaker
4	Automated rollback and human takeover
5	Blast-radius isolation and recovery drills

Assurance and Evaluation

Score	Description
0	No evaluation
1	Manual testing before release
2	Basic prompt or output evaluation
3	Scenario evaluation and red team testing
4	Runtime evaluation and drift detection
5	Continuous assurance across policy, behavior, security, and outcomes

Appendix C. Architecture Review Packet

- Agent profile and business context.
- Agent Bill of Materials defining the agentic system boundary.
- Consumer population and Control Responsibility per gravity dimension.
- AFI, EGI, SMI, and orbit classification.
- Non-compensatory rule outcome: which rules fired, on which dimensions, and the resulting ceiling.
- Deployment zone and deployment decision.
- Architecture diagram.
- Tool/MCP inventory and data-source inventory, including read/write capability and reversibility.
- Identity, authorization, policy, observability, recovery, and assurance models.
- Threat model and mitigation plan.
- Evidence tier summary.
- Open control gaps and control uplift plan.
- Risk acceptance record if applicable.
- Reassessment date and triggers.

Appendix D. References

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